Enrollment No._____

GANPAT UNIVERSITY

B.Sc. (IT) SEM-I EXAMINATION

U21A3DM: INTRODUCTION TO DATA ANALYSIS AND DATA MANAGEMENT

Time: 3 Hours] **[Total Marks: 60**

JANUARY - JUNE 2020

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following: (ANY SIX OUT OF EIGHT) (30)

A. Explain formatting documents with figure in detail.

B. What is mail merge? Write down its steps with all options.

C. Write down any five date & time function.

D. List out types of charts. Write down the steps for creating charts.

E. Discuss the menus of math with figure.

F. Write down the steps to create new database.

G. What is IMPRESS? Write down the advantages of IMPRESS.

H. Write a short note on Master Templates.

2. Answer the following: (ANY TWO) (09)

A. What is Style? Describe different styles available in writer.

B. Write a short note on printing documents.

C. Write a short note on drawing tools.

3. Answer the following: (ANY TWO) (09)

A. Describe how to work with multiple sheets?

B. Write a short note on formula editor in math.

C. Explain the toolbars of math in detail.

4. Answer the following: (ANY ONE) (06)

A. What is Database? List out the advantages of DATABASE.

B. Define the term Tables. Write down its steps to create it.

5. Answer the following: (ANY ONE) (06)

A. What is presentation? Write down the steps to create new presentation.

B. Write a short note on re-usability in Impress.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. (IT) SEM-I EXAMINATION (CBCS)

U21A2DEL: INTRODUCTION TO DIGITAL ELECTRONICS

Time: 3 Hours] JANUARY - JUNE 2020 [Total Marks: 60

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

Q-1. Answer the following: (Any Six) (30)

I. List out types of number systems. Explain octal number system in brief.

II. What is parity bit? Explain checksum method in brief.

III. What is universal gate? Explain all universal gates in details.

IV. State and prove the Demorgan's theorems.

V. Add -17 and -30 with 2's complement using 8 bits.

VI. Discuss 4 to 1 multiplexer in brief.

VII. Explain ring counter in details.

VIII. Write a short note on types of RAM in brief.

Q-2. Answer the following: (Any Two) (08)

I. Discuss error correcting code (Hamming code) method in brief.

II. Explain odd and even parity code methods with suitable example.

III. [A] $(54.31)_8 \rightarrow (\ ?\ )_{16}$ [B] $(127.5625)_{10} \rightarrow (\ ?\ )_2$

Q-3. Answer the following: (Any Two) (08)

I. What is logic gate? Explain all basic gates in brief.

II. Implement basic gates using any one universal gate.

III. Simplify following Boolean expression using K-Map:

$F(A,B,C,D) = \Sigma(0,1,4,5,7,8,9,12,13,15)$.

Q-4. Answer the following: (Any One) (06)

I. Explain full adder with its diagram.

II. Discuss 4 to 16 Decoder in brief.

Q-5. Answer the following: (Any One) (04)

I. List out the types of ROM. Explain ROM in details.

II. Write a short note on NAND latch.

Q-6. Answer the following: (Any One) (04)

I. Explain shift left register in brief.

II. Write a short note on ripple counter.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY
B.Sc. (IT) SEM-III EXAMINATION

SUBJECT CODE: U23A1DFS **SUBJECT NAME: DATA STRUCTURE**

Time: 3 Hours] **[Total Marks: 60**

JANUARY - JUNE 2020

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. **Answer the following: (ANY SIX OUT OF EIGHT)** **(30)**

A. What is Linear Data Structures? Explain with figure.

B. What is difference between primitive and Non primitive Data Structures?

C. Write an algorithm for the operations on stack.

D. Define a linked list. Write the advantage and disadvantage of lined list?

E. Define a circular linked list? Write an algorithm to invert a circular linked list.

F. What is tree? Explain the following terms in tree terminology: a Leaf node b. Path c. Siblings

G. Write the brief note on AVL Tree.

H. Sort a following list of elements using Quick sort:

65 70 75 80 85 60 55 50 45

2. **Answer the following: (ANY ONE)** **(5)**

A. Define a Data Structure? Explain types of data structures.

B. What is Non-Linear Data Structures? Explain with figure.

3. **Answer the following: (ANY Two)** **(5+5)**

A. Write an algorithm for operations of Queue.

B. What is Sparse Matrix? Represent an given Array A in sparse matrix.

$$A=\begin{pmatrix} 0 & 0 & 0 & 0 & 9 & 0 \\ 0 & 8 & 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 \\ 0 & 0 & 2 & 0 & 0 & 0 \end{pmatrix}$$

C. Write an algorithm to insertion of a node into a doubly linked list at the beginning.

4. **Answer the following: (ANY TWO)** (5+5)

A. Explain the Binary Tree Traversals.

B. Explain the Breadth-First Search Traversal Algorithms also write the algorithms.

C. Define a graph and explain its operations.

5. **Answer the following: (ANY ONE)** (5)

A Define Insertion sort. Sort a following list of elements using Insertion sort: elements of the array are : 8,4,5,3,2,9,4,1

B Write an algorithm for binary search.

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY
B.Sc. (IT) SEM-V EXAMINATION

U25A1WPF: (ELECTIVE-II) WEB PROGRAMMING FRAMEWORK

Time: 3 Hours] **[Total Marks: 60**

JANUARY - JUNE 2020

Instructions:
1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following: (ANY SIX OUT OF EIGHT) (30)

A. What is Magento? Explain Customer Journey for any ecommerce website.

B. Explain in brief : Admin Store Panel

C. List out types of products in magento and explain any 2 in details.

D. Explain Catalog Price Rules with suitable example.

E. Explain Free shipping and Flat rate shipping method in detail.

F. Explain Check / Money Order and Bank Transfer payment methods in brief.

G. Explain Marketing Reports and Review Reports in brief.

H. Write a Sort note on : Widgets

2. Answer the following: (ANY TWO) (08)

A. Explain the terms: Logo, Favicon, Store Demo Notice.

B. Explain Website, Store and Store View in brief.

C. Write down the steps for configuring: Store Information, Currency setup in magento.

3. Answer the following: (ANY TWO) (08)

A. What is Attribute Set? How to add attributes in attribute set?

B. Explain Up-sell, Cross-sell and Related products in Magento.

C. How to Manage Advance Pricing in Magento?

4. **Answer the following: (ANY ONE)**

A. Explain Table rate shipping method in detail.

OR

A. Write a short note on: Invoices

B. Define the term: Shipping Label (02)

5. **Answer the following: (ANY ONE)** (05)

A. Write a short note on : Themes

OR

A. Write a Sort note on: Pages

B Define the term: Customer Groups (02)

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. (IT) SEM-V EXAMINATION

U25A30PS: OPERATING SYSTEM

[Time: 3 Hours] [Total Marks: 60]

JANUARY - JUNE 2020

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following: (ANY SIX OUT OF EIGHT) **(30)**

A. Discuss time sharing operating system.

B. Discuss real time system and differences among hard and soft real time system.

C. Discuss virtual machine architecture along with their advantages and disadvantages.

D. Discuss deadlock and necessary conditions for deadlock occurrence.

E. Discuss shortest job first process scheduling algorithm with suitable example.

F. Discuss critical section problem and how can we handle it with semaphores synchronization mechanism.

G. Discuss segmentation in memory management.

H. Discuss monolithic kernel system architecture. What are advantages of monolithic architecture over other architectures.

2. Answer the following: (ANY FOUR) **(8)**

A. Discuss comparison between multiprogramming and multiprocessing operating system and prove all multiprocessing systems are multiprogramming but all multiprogramming OS are not necessarily multiprocessing OS.

B. Discuss buffering and SPOOLING concepts.

C. Discuss interrupt, its functions and interrupt handling mechanisms.

D. Discuss Direct Memory Access (DMA) architecture.

E. Discuss hierarchical memory architecture and advantages of caching in a computer system.

3. Answer the following: (ANY TWO) **(6)**

A. Discuss different states of a process? Describe them with suitable diagram.

B. Discuss Process control block and its different attributes.

C. Discuss deadlock avoidance with Banker's algorithm.

4. Answer the following: (ANY TWO) **(6)**

A. Discuss logical and physical address space.

B. Discuss concept of swapping in memory management.

C. Discuss concept of internal and external fragmentation.

5. Answer the following: (ANY FOUR) **(10)**

A. Describe difference between *cmp* and *diff* commands.

B. Describe *grep* command.

C. Describe difference between pipe and redirection directives.

D. Describe file permissions types and their handling using *chmod* command.

E. Name any two shells of UNIX operating system.

F. Describe *ln* command.

END OF PAPER

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